Grouping and Functions

Python course - Lesson 4+5

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Previous Lesson Topics

- NumPy
  - NumPy array
  - NumPy matrix
- Pandas
  - Pandas Series
  - Pandas DataFrame

Today's Topics

- Group and aggregate
- Pivot tables
- User functions
- Lambda functions
- Basic visualizations

Group and Aggregate

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Groupby() Method

df.groupby ('column_name')

df.groupby (['column_name1', 'column_name2' …])

Creates a group object based on common values in the selected column

Groupby() Method

Method	Action
size ()	Compute group sizes
first ()	Compute first of group values
last ()	Compute last of group values
get_group ('col')	Construct DataFrame from group with provided name
g['col'].max ()	Compute max of group values
g['col'].min ()	Compute min of group values
g['col'].mean ()	Compute mean of groups, excluding missing values
g['col'].sum ()	Compute sum of group values
g['col'].describe ()	Generate descriptive statistics

Group Aggregations

```
Group_object .agg ( {'col_1' : ['agg_func1', 'agg_func2',...] ,
                     'col_2' : ['agg_func1', 'agg_func2',...]
                       . . .
                     })
```

Aggregate using one or more operations over the specified axis

Group Aggregations

Method	Action
min ()	Compute max of group values
max ()	Compute min of group values
mean ()	Compute mean of groups, excluding missing values
sum ()	Compute sum of group values
count ()	Compute count of values in the group

Pivot Table

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Pivot Table

df.pivot_table (parameters)

- Used for pivoting with aggregation of numeric data
- Can be used to create spreadsheet-style pivot tables
- Inside the values can use aggregate function

Pivot Table – Example

df.pivot_table (values='B', index='A' , columns='C', aggfunc='count')

	A	B	C
0	1	B0	BIG
1	1	B1	BIG
2	2	B2	SMALL
3	2	B3	SMALL
4	2	B4	SMALL
5	1	B5	SMALL
6	1	B6	BIG

	C	
	BIG	SMALL
1	3	1
2	0	3

Pivot Tables Parameters

df.pivot_table (parameters)

Parameter	Action
values = 'column_name'	Column or a list of columns to aggregate
Index = 'column_name'	Column to group by on the pivot table index
columns = 'column_name'	Column to group by on the pivot table column
aggfunc = 'aggregate function'	Function to use for aggregation

User Define Functions

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User Define Function

```
def func_name (prm_name1,...):
    <function body>
        <return value>
```

Tab

Lambda Functions

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Lambda Function

Func_name = lambda prm1,... : <return value>

Lambda function is anonymous function which involve small and concise syntax

Local & Global Parameters

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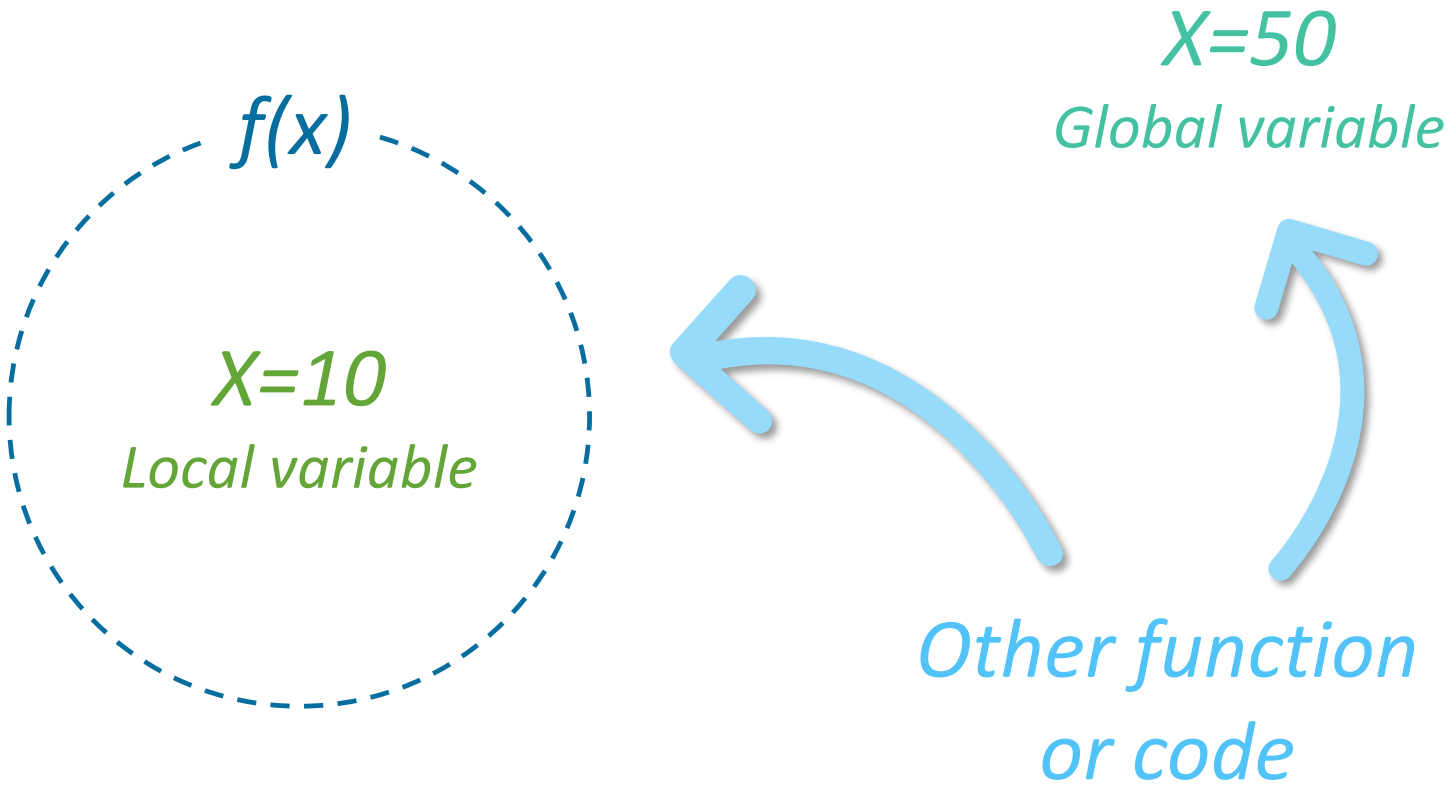
Variable Scope

difference between local and global variable:



Variable Scope

difference between local and global variable:

- when x is out of the function - its global variable
- when x is defined in a function - its local variable

(even if we have global variable with the same name)

Thank You

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